

Speed of light

The speed of light in free space is an important and intriguing constant of nature. Whether the light comes from a laser on a desktop or from a distant galaxy that is hurtling away at fantastic speeds, the measurement of the speed of light will yield the same constant value. In more precise terminology, the speed of light is independent of the relative velocities of the light source and the observer.

As Einstein first presented in his special theory of relativity, the speed of light is critically important in some surprising ways:

1. The speed of light establishes an upper limit to the speed that may be imparted to any material object.
2. Objects moving near the speed of light follow a set of physical laws drastically different, not only from Newton's Laws, but from the basic assumptions of human intuition.

It is not surprising that a great deal of time and effort has been invested in measuring the speed of light. Some of the most accurate measurements were made by Albert Michelson between 1926 and 1929. Michelson measured the speed of light in air to be 2.99712×10^8 m/s. From this result he deduced the speed of light in free space to be 2.99796×10^8 m/s. Today, the speed of light in free space is, by definition, $(2.997924562 \pm 0.000000011) \times 10^8$ m/s exactly.

Our approach to the measurement of the speed of light will be direct and intuitive, and simpler than Michelson's experiment. We will use a diode laser as our light source. We will send pulses of light across the lab, measure how long they take to come back, and deduce the speed of light from the knowledge that light travels at constant speed.

Apparatus

- Diode Laser — The diode laser emits an intense, narrowly- focused beam of light at a wavelength of approximately 650nm. In this experiment, the diode laser is powered by a function generator, which modulates the light intensity at about 4 MHz. The laser is equipped with adjustment screws for precisely aiming the light at the mirror.
- Function Generator — Provides power to the diode laser. The laser is turned on by a DC offset voltage from the function generator and modulated by a superimposed sine wave having frequency of about 4 MHz.
- Concave Mirror — The concave surface of the mirror helps to focus the light as it is reflected. The mirror is also equipped with adjustment screws for aiming the light back to the receiver.
- Converging Lens — The lens is used to focus returning light onto the active element of the light receiver. The lens has a focal length of 12.7cm.
- Light Receiver — The receiver is designed for receiving audio and video signals transmitted via modulated light. Since the light receiver is sensitive to very high-frequency modulation, it is ideally suited to this experiment. There are two sensitive elements on the receiver; in this experiment, you will use only the one labeled "Video" (which is sensitive to high frequency signals). (The "Audio" receiver, which is sensitive to lower frequency signals, will not be used in this experiment.)
- Oscilloscope — The oscilloscope is used to view both the modulating signal and the returned signal simultaneously. The phase difference between these two will be converted to a time delay.

- Other Equipment — Pasco OS-9172 Alignment bench, cables, tape measure.

Laser light safety notes

We will use a relatively safe, low power laser. Nevertheless, take the following precautions:

1. Never look directly into the laser beam, either directly, or as it is reflected from a mirror or other surface.
2. Keep the optical path at floor level and keep your eyes well above floor level at all times.
3. Only people involved with the experiment should be nearby.

Experimental setup

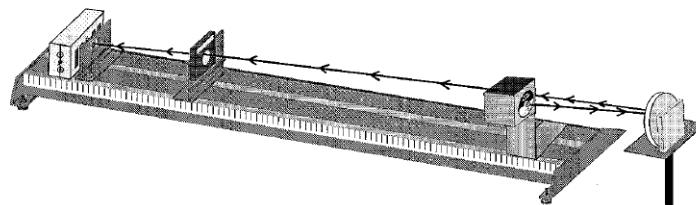


Figure 1: Experimental set-up showing the laser, concave mirror, lens, and receiver. The laser, lens, and receiver are mounted on the laser alignment bench. The mirror is shown mounted on a stand, but in our arrangement both the bench and the mirror will be placed directly on the laboratory floor.

1. Place the laser, lens, and receiver on the alignment bench as shown in the figure above.
2. Place the bench on the floor, making sure that you have 12-15 meters of available space in front of it.
3. Connect your function generator to the diode's input power and to channel 1 of the oscilloscope. This will be the signal you trigger from.
4. Connect the video output of the light receiver to channel 2 of the oscilloscope.
5. Set the function generator for a sine wave with a frequency as close to 4 MHz as you can make it, and press the DC Offset button to turn the DC offset function on. Turn the Output and DC Offset knobs completely counterclockwise to the zero position.
6. Power on the function generator and turn up the DC Offset knob until you see laser light.
7. Align the laser so that it propagates along a line on the floor and that its height is constant throughout. It will make alignment easier afterwards. Once this alignment is done,
8. Place the mirror approximately 1 m in front of the laser and use it to reflect the beam (through the lens) on the light receiver sensor. Make sure you go through the center of the lens. It will make the procedure simpler.
9. Check the strength of the signal on the oscilloscope. You want to see a sine wave with decent amplitude (signal-to-noise). You might have to play with the function generator some more (amplitude and DC offset) to get a better signal. Set the oscilloscope to average 128 cycles.

Experimental setup

- Using a tape measure, record the total path length of the light pulses.
- Using the cursors on the oscilloscope, measure the time delay between the two traces.
- Also using the cursors, also record your uncertainty in the time delay.
- Move the mirror back about 60 cm and repeat the measurement.

Experimental setup

Make a graph of the time delay as a function of round trip distance. Find the slope of this graph and its uncertainty. Convert this slope (and uncertainty) into a value for the speed of light (and uncertainty).